

Design and Testing of the PAX Thermo-electric Cooling System

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Introduction

The thermo-electric cooler (TEC) for the PAX is driven by an electronics board which is located in the rim box. The TEC can either cool or heat the PAX. The PAX temperature is measured by a thermistor and is maintained at a constant 20 C using proportional and integral feedback. The 20 C setpoint is determined by fixed resistors on the drive board, but could be changed to a different fixed temperature. The setpoint can also made controllable by varying an analog voltage input to the driver board.

The TEC drive board also provides 4 analog voltage outputs, which are a measure of the temperatures of 4 thermistor temperature sensors. The mounting locations for the 4 thermistors are: 1, on the bottom of the PAX, 2, on one TEC heatsink, 3, in the cooling airstream upstream from the TEC heatsinks, and 4, on the rim box heatsink to which the TEC drive electronics board (and most other rim box electronics) is mounted.

The Thermistors and Temperature Monitoring

The thermistor used is the Epcos model B57703M0103G040 thermistor which has a nominal resistance of 10k ohms at 25 C, and a relatively large $\beta = 3988$ K, giving it high sensitivity. It is packaged in a wire terminal lug, ready for mounting. The resistance tolerance for the thermistor is $\pm 2\%$, which equates to a temperature tolerance of ± 0.43 C at 20 C. The accuracy of the drive electronics board is about 20 times better than the thermistor, so the overall accuracy of the thermistor plus its drive electronics is better than 0.5 C; also, swapping drive boards will not affect the reading from a particular thermistor significantly.

The output voltage range for the TEC driver board temperature sensors is chosen to be 0 to +5.0 V, to match the range of the 10 bit ADC located in the rim box. For the PAX box temperature sensor, which should regulate at 20 C, the electrical gain is chosen so that a PAX temperature of approximately 25 to 15 C will produce the full 0 to +5 V range of the ADC. Temperatures outside this range will saturate the output voltage at either 0 or 5 V. For a 10 bit ADC this gives a quantization of 10 mK per bit. The following expressions should be used to calculate the PAX temperature from the ADC voltage:

$$V_{\text{thermistor}} (V) = 0.059074 * V_{\text{ADC}} (V) + 1.1044$$

$$R_{\text{thermistor}} (\text{kohms}) = 11.372 / \{ (2.3815/V_{\text{thermistor}}) - 1 \}$$

$$T (K) = 298.15 / \{ 1 + (298.15/3988) * \ln(R_{\text{thermistor}}/10.0) \}$$

$$T (C) = T (K) - 273.15$$

These expressions are "exact", and result from the theoretical characterization of the thermistor and the gain of the electronic circuit.

For the other 3 sensors, which measure the TEC heatsink, the PAX airstream, and rim box heatsink, the anticipated temperature swings are much larger. The electronical gains are chosen so that +50 C to -10 C corresponds to the 0 to +5 V ADC range. For a 10 bit ADC, this gives a quantization of about 50 mK per bit near the middle of the temperature range, rising rather steeply near the extremes of the range to near 80 mK per bit. The following expressions should be used to calculate temperatures from these 3 ADC voltages:

$$V_{\text{thermistor}} (V) = 0.60243 * V_{\text{ADC}} (V) + 1.2952$$

$$R_{\text{thermistor}} (\text{kohms}) = 10.0 / \{ (5.0/V_{\text{thermistor}}) - 1 \}$$

$$T (K) = 298.15 / \{ 1 + (298.15/3988) * \ln(R_{\text{thermistor}}/10.0) \}$$

$$T (C) = T (K) - 273.15$$

The Thermo-Electric Cooling System

Heat is removed from or added to the PAX box by a pair of Marlow Industries model DT12-4-01 TECs, each sandwiched between the bottom plate of the PAX box and a 10 inch length of Aavid Thermalloy heatsink extrusion 74765. The antenna cooling air passes through the 2 heatsinks (in parallel) on its way to the frontend cryocooler. At a flow rate of 13 CFM, each heatsink has a thermal resistance of 0.2 C/W. The PAX box is well insulated from both the ambient air and the cooling air which flows through the heatsinks. Some of the insulation also functions to seal off unwanted airflow paths, causing all the cooling air to flow through the heatsinks.

The PAX cooling system is designed for the conditions found on a hot summer afternoon at Hat Creek, and has a substantial excess capacity under those conditions. The design condition assumes that the ambient air temperature outside the PAX case is 40 C, the antenna cooling air arrives at the TEC heatsinks at 30 C, and the PAX box is 20 C. Under these conditions, the heat leak to the PAX from ambient is 4.25 W and the heat leak from the TEC heatsinks is 1.15 W. These heat leaks through the insulation, plus a

0.4 W heat leak through the screws which mount the TEC/heatsink assemblies, and an assumed 12 W maximum dissipation in the PAX, give a total heat load of 17.8 W to be removed by the TECs. This load is shared by the 2 TECs, resulting in a cold end heat load of 9 W each. For this heat flow, the temperature drop in the aluminum spacer block is 0.9 C, and the temperature drop in the 2 interfaces (using Marlow thermal grease) is estimated to be 1.1 C, giving a TEC cold end temperature of 18 C. The electrical power into each TEC needed to lift 9 W across a 15 C temperature difference (18 C to 33C) is about 6 W (1.2 A at about 5 V). Thus, the heat flow at each TEC hot end is 15 W. The thermal resistance of the heatsink is 0.2 C/W giving a heatsink temperature 3 C hotter than the airstream, or 33 C.

The TEC drive circuit operates from a single +24.0 V supply (the voltage used for driving the frontend cryocooler), and produces a differential output of either polarity, centered at 12 V, to provide either cooling or heating of the PAX. The 2 TECs are wired in series, resulting in an input impedance of about 8 to 10 ohms, depending on temperatures and heat flow. When cooling the PAX, the maximum current to the TECs is limited to 2.0 A using a current limiting feature on the OPA548 power op amps (this limit is enforced separately on each output since they both float). This limits the maximum electrical power into the TECs to about 36W. When heating the PAX, the maximum TEC voltage is limited to about -10.5 V, corresponding to an electrical power of about 12 W. The OPA548 op amps also feature built-in over-temperature protection.

The TEC drive electronics uses proportional and integral feedback (the circuit board is actually laid out for PID control, but the components for differential feedback are not installed). The Zeigler-Nichols method was used to find appropriate gains under a typical operating condition (15 W removed from the PAX, the PAX regulating at 20 C, and cooling air entering at 30 C).

Testing

The temperature servo is well behaved over a wide range of temperatures and heat loads. The stability of the control loop was checked for a range of heat loads (from about -7 to 23 W removed from the PAX) in combination with a range of cooling air temperature (from 0 C to 35 C). The system was found to be well behaved in all cases checked. In response to a transient, with P or PI control the temperature of the PAX box typically oscillates with a period of about 85 seconds. For all load/temperature conditions tested, the amplitude of the temperature oscillation was damped by a factor of 3 in about 4 to 7 cycles. The slowest decay of oscillations occurred at low cooling air temperature and low PAX dissipation, where the TEC heated during part of the oscillation cycle and cooled during the other part of the cycle.

summary.txt

summary.txt

T_{pax} 20 C
 dT_{grease} 1.1 C guess
 dT_{spacer} 0.9 C
 deltaT spacer = 10W * .865in / (1.5in)² / 168W/m/K / .0254m/in = 0.90 C
 T_{teccold} 18 C
 T_{coolingairin} 30 C
 dT_{heatsink} 3 C www.aavidthermalloy.com extrusion 74765
 13CFM / (2.341/144 ft²) = 800 ft/min, 10 in length -> R = 0.2 C/W
 T_{techot} 33 C
 dT_{tec} 15 C = 33 - 18
 T_{ambient} 40 C
 A_{insulhtsink} 83.8 in² A = 2*4.24*9.88in²
 A_{insulambient} 348.3 in² A = 2*12.5*13+51*2.1 - 83.8 in²
 Q_{htsink} 1.15 W
 Q = .25 BTUh*in/ft²/F * 9F/5C * 1W/3.414BTUh * 1ft²/144in² *
 15C / 1in * 83.8 in² = 1.15 W
 Q_{ambient} 4.25 W
 Q = .25 BTUh*in/ft²/F * 9F/5C * 1W/3.414BTUh * 1ft²/144in² *
 20C / 1.5in * 348.3 in² = 4.25 W
 Q_{insulttotal} 5.4 W
 Q_{dissipated} 12 W max Ipax = A, Vpax = V
 Q_{screws} 0.4 W
 #6-32 screw: qdot = 15 W/m*K * 25K/1.25in * pi/4 * .13in² * .0254m/in = .101 W
 Q_{cold} 18 W 12 + 5.4 + .4 = 17.8
 (9 W per tec)
 Q_{hot} 15 W per tec = 9W + 6W
 V_{tec} 4.8 V
 I_{tec} 1.2 A
 dT_{max} 64 C
 dT/dT_{max} .234
 optimal Q/Q_{max} .18 optimal Q_{max} = 9/.18 = 50 W
 max Q/Q_{max} .77 minimum Q_{max} = 9/.77 = 12 W
 Q_{max} for DT12-4 36 W

dT/dt/dQ 0.9 C/hr/W for 1W excess power, PAX temp will rise at approx,
 $R = 1 \text{ W} / (.90 \text{ J/g/C} * 2.7 \text{ g/cm}^3 * (2.54 \text{ cm/in})^3 * 12.5 * 13 * .6 \text{ in}^3) =$
 $1 \text{ J/s} / (.90 \text{ J/g/C} * 4300 \text{ g}) = 0.00026 \text{ C/s} = 0.93 \text{ C/hr}$

dT/dQ 0.6 C/square/W for 1W load on pax bottom,
 $1\text{W} = 168 * .38 * 1/1 * .0254 * dT, dT = .92 \text{ C}$

Test results

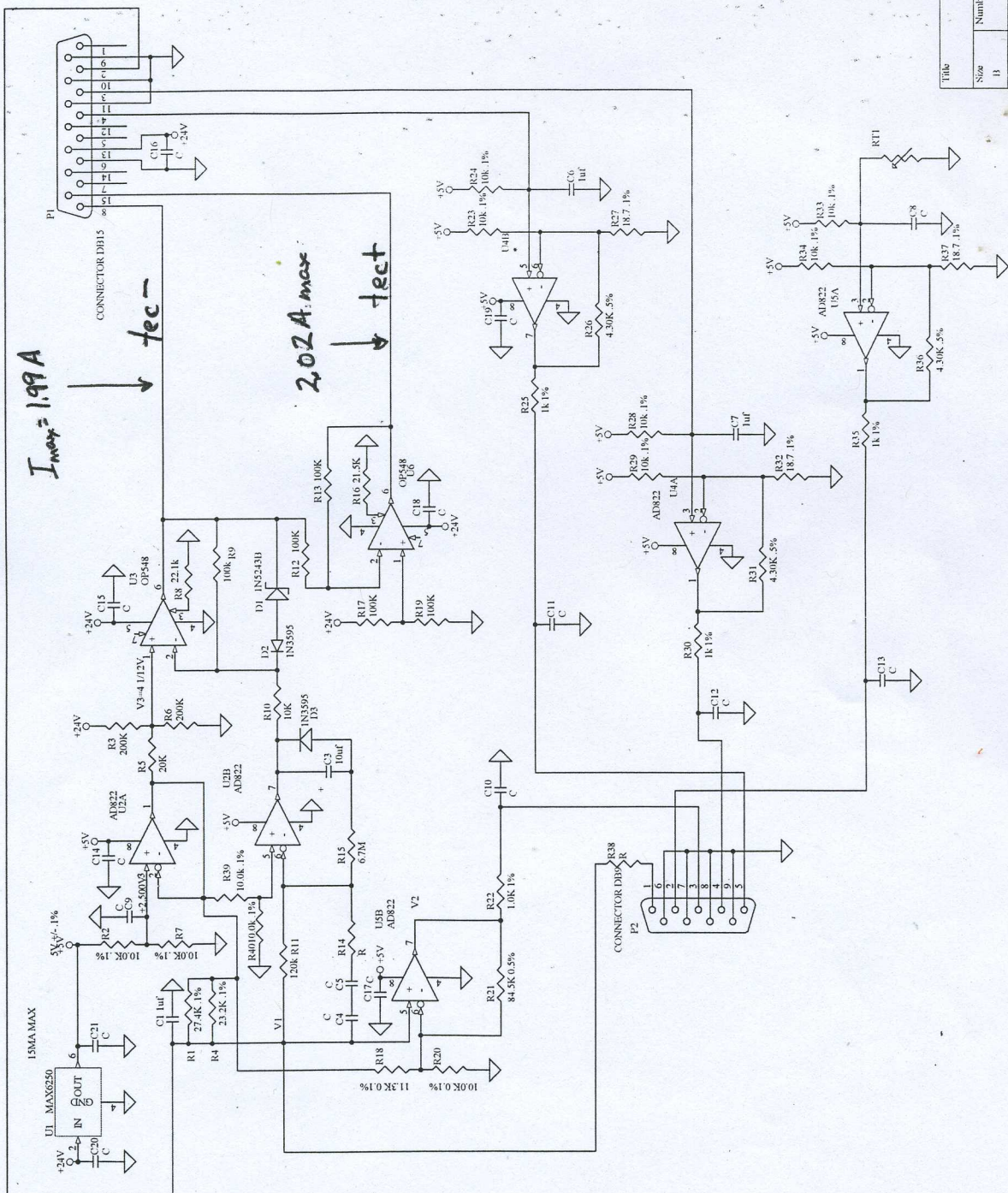
T _{pax}	T _{amb}	dT	PAX Pload	Vtec	from Marlow curves 2*Ptec@dTtotal
21.5	30	8.5	18+3	5.1	21 @ 17
21.5	30	8.5	15+3	4.8~	18 @ 17
21.5	0	-21.5	15-6	-2.9	
21.5	15	-6.5	12-3	-0.1~	7 @ -
21.5	10	-11.5	8-3	0	5 @ -8 guess
15	5	-10		0	

Pload is Phtr_{pax} + calculated insulation leak
 2*Ptec is cold heat lift from Marlow data curves
 data suggests dT_{total} is about dT+8 degrees

wire DT12-4 tecs in series; expect max cooling 2.0 A, 16 - 20 V (9 ohms, <40W continuous)
 max heating 0.6 A, ~~4-8 V~~ (10 ohms)

~1.0 A

2.4 max cool
 - 10.5 V max heat



Title			
TEC DRIVER			
Size	Number	Revision	
B	1034		
Date	8/9/2006	Sheet of	
File	P:\Work\Jagadech\work\1034 Sch.dwg		
	Drawn By:		

/ataheat/tecwiringlist.txt

TEC driver card, linear version

Input/output list

Connector 1 - 15 pin sub D

female on board, male on cable going to power supply and PAX Case cable
all cable wires 24 ga except as noted (20 ga is 0.01 ohm/ft)

pin	function
1	NC or shield
2	PAX thermistor +
9	PAX thermistor -
3	airstream (aux1) thermistor +
10	airstream (aux1) thermistor -
4	TEC heatsink (aux2) thermistor +
11	TEC heatsink (aux2) thermistor -
6	supply ground (20 ga wire in cable)
13	supply +24 V (20 ga wire in cable)
8	TEC drive - (20 ga wire in cable)
15	TEC drive + (20 ga wire in cable)

Connector 2 - 9 pin sub D

female on board, male on cable to Rim Box controller board

pin	function
1	setpoint voltage
2	rimbox heatsink thermistor monitor
6	ground
3	PAX thermistor monitor
7	ground
4	airstream (aux1) thermistor monitor
8	ground
5	TEC heatsink (aux2) thermistor monitor
9	ground

Applications

- Surface temperature measurement, e.g. on housings and pipes

Features

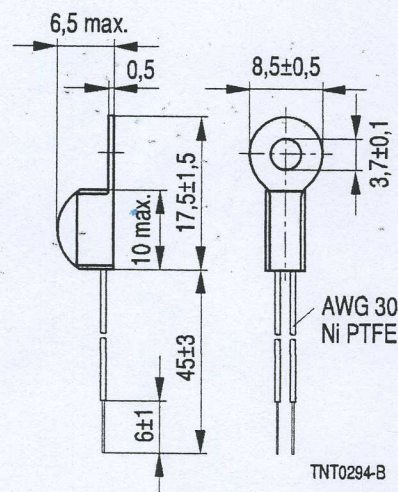
- High accuracy
- Easy mounting
- Good thermal coupling through metal tag
- Thermistor encapsulated in metal-tag case
- PTFE-insulated leads of silver-plated nickel wire, AWG 30

Options

Alternative resistance ratings, rated temperatures, resistance tolerances and lead lengths available on request

Delivery mode

Bulk



Dimensions in mm
Approx. weight 0,8 g

Climatic category (IEC 60068-1)		55/125/56	
Max. power at 25 °C	P_{25}	150	mW
Resistance tolerance	$\Delta R_N / R_N$	± 2 %	
Rated temperature	T_N	25	°C
B value tolerance	$\Delta B / B$	± 1 %	
Dissipation factor (in air)	$\delta_{th}^{1)}$	approx. 2,6	mW/K
Thermal cooling time constant (in air)	$\tau_c^{1)}$	approx. 28	s
Heat capacity	$C_{th}^{1)}$	approx. 73	mJ/K
Test voltage ($t = 1$ s)	V_T	1	kVAC

R_{25}	No. of R/T characteristic	$B_{25/100}$	Ordering code
Ω		K	
5 k	8016	3988	B57703M0502G040
10 k	8016	3988	B57703M0103G040
30 k	8018	3964	B57703M0303G040

1) Depends on mounting situation

Reliability data

Test	Standard	Test conditions	$\Delta R_{25}/R_{25}$ (typical)	Remarks
Storage in dry heat	IEC 60068-2-2	Storage at upper category temperature $T: 125\text{ °C}$ $t: 1000\text{ h}$	< 1 %	No visible damage
Storage in damp heat, steady state	IEC 60068-2-3	Temperature of air: 40 °C Relative humidity of air: 93 % Duration: 56 days	< 1 %	No visible damage
Rapid temperature cycling	IEC 60068-2-14	Lower test temperature: -55 °C Upper test temperature: 125 °C Number of cycles: 10	< 0,5 %	No visible damage
Endurance		$P_{\max}: 150\text{ mW}$ $t: 1000\text{ h}$	< 1 %	No visible damage
Long-term stability (empirical value)		$T: 70\text{ °C}$ $t: 10\,000\text{ h}$	< 2 %	No visible damage

PAX thermistor

thermistor1.xls

gives resistance vs. temp curves for thermistors with various beta
 this info from <http://www.thermometrics.com/assets/images/ntcnnotes.pdf>

in practice observe approximately straight line relation between $\ln(R)$ and $1/T(K)$

$$R(T) = R(T_0) * \exp(\beta(T_0 - T)/T/T_0)$$

if define $\alpha = (1/R) * (dR/dT)$, then $\alpha = -\beta/T^2$

$$T(K) = T(C) + 273.15$$

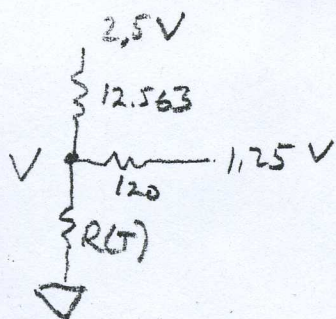
T_0	(C)	25		R2	
beta	(K)	3988	Epcos	11.3	R4
R(T_0)	(kohm)	10		R3	84.5
Tstart	(C)	14		10.0	
Tstep	(C)	0.5			
Rseries	(kohm)	12.563			
Vbias	(V)	2.5			
Vadc,f.s.	(V)	5			
adc bits		10			

(over)

incl. 120k tied to virtual 1.25V

T (C)	R (kohm)	alpha	Vtherm (V)	Vadc (V)	Tresol (K/bit)	V _{therm}	V _{adc}
14.0	16.693	-0.04837	1.4265	5.452			
14.5	16.295	-0.04820	1.4116	5.201	-0.00974	1.3887	4.8132
15.0	15.907	-0.04803	1.3968	4.951			
15.5	15.531	-0.04786	1.3820	4.700			
16.0	15.164	-0.04770	1.3673	4.450	-0.00976		
16.5	14.807	-0.04753	1.3525	4.200			
17.0	14.460	-0.04737	1.3377	3.951			
17.5	14.122	-0.04721	1.3230	3.701			
18.0	13.793	-0.04705	1.3083	3.453	-0.00983		
18.5	13.473	-0.04688	1.2937	3.205			
19.0	13.161	-0.04672	1.2791	2.957			
19.5	12.858	-0.04656	1.2645	2.711			
20.0	12.563	-0.04641	1.2500	2.465	-0.00995	1.2500	2.4652
20.5	12.275	-0.04625	1.2355	2.220			
21.0	11.995	-0.04609	1.2211	1.976			
21.5	11.722	-0.04593	1.2067	1.732			
22.0	11.456	-0.04578	1.1924	1.490	-0.01010		
22.5	11.197	-0.04562	1.1782	1.249			
23.0	10.945	-0.04547	1.1640	1.009			
23.5	10.700	-0.04532	1.1499	0.770			
24.0	10.460	-0.04517	1.1358	0.533	-0.01030		
24.5	10.227	-0.04501	1.1219	0.296			
25.0	10.000	-0.04486	1.1080	0.062		1.1143	0.1681
25.5	9.779	-0.04471	1.0942	-0.172	-0.01049		
26.0	9.563	-0.04456	1.0805	-0.404			

 $\Delta V' = \Delta V \cdot 0.9445$
 $\Delta V' = \Delta V \cdot 0.9559$



$$\frac{2.5 - V}{12.563} + \frac{0 - V}{R_{LT}} + \frac{1.25 - V}{120.0} = 0$$

$$V \left(\frac{1}{12.563} + \frac{1}{R_{LT}} + \frac{1}{120.0} \right) = \frac{2.5}{12.563} + \frac{1.25}{120}$$

w/ fuels

3 full

2 X only

1 Y only

2 unknown

Aux 1, Aux 2, heatsink thermistors

thermistor2.xls

gives resistance vs. temp curves for thermistors with various beta
this info from <http://www.thermometrics.com/assets/images/ntcnotes.pdf>

in practice observe approximately straight line relation between $\ln(R)$ and $1/T(K)$

$$R(T) = R(T_0) * \exp(\beta * (T_0 - T) / T * T_0)$$

if define $\alpha = (1/R) * (dR/dT)$, then $\alpha = -\beta / T^2$

$$T(K) = T(C) + 273.15$$

T_0	(C)	25		R2	
beta	(K)	3988	Epcos	10.0	R4
R(T_0)	(kohm)	10		R3	4.3
Tstart	(C)	-10		18.7	
Tstep	(C)	2.5			
Rseries	(kohm)	10			
Vbias	(V)	5			
Vadc,f.s.	(V)	5			
adc bits		10			

T	R	alpha	Vtherm	Vadc	Tresol
(C)	(kohm)		(V)	(V)	(K/bit)
-10.0	59.241	-0.05759	4.2779	4.951	
-7.5	51.367	-0.05651	4.1852	4.797	-0.07634
-5.0	44.658	-0.05546	4.0852	4.631	
-2.5	38.926	-0.05444	3.9781	4.453	
0.0	34.015	-0.05345	3.8640	4.264	-0.06273
2.5	29.796	-0.05249	3.7436	4.064	
5.0	26.163	-0.05155	3.6174	3.855	
7.5	23.026	-0.05063	3.4861	3.637	
10.0	20.311	-0.04974	3.3504	3.412	-0.05356
12.5	17.956	-0.04887	3.2115	3.181	
15.0	15.907	-0.04803	3.0700	2.946	
17.5	14.122	-0.04721	2.9272	2.709	
20.0	12.563	-0.04641	2.7839	2.471	-0.05143
22.5	11.197	-0.04562	2.6412	2.234	
25.0	10.000	-0.04486	2.5000	2.000	
27.5	8.947	-0.04412	2.3611	1.769	
30.0	8.020	-0.04340	2.2253	1.544	-0.05493
32.5	7.202	-0.04269	2.0934	1.325	
35.0	6.479	-0.04200	1.9658	1.113	
37.5	5.838	-0.04132	1.8430	0.909	
40.0	5.269	-0.04067	1.7254	0.714	-0.06403
42.5	4.764	-0.04003	1.6133	0.528	
45.0	4.313	-0.03940	1.5068	0.351	
47.5	3.912	-0.03879	1.4059	0.184	-0.07504
50.0	3.553	-0.03819	1.3108	0.026	



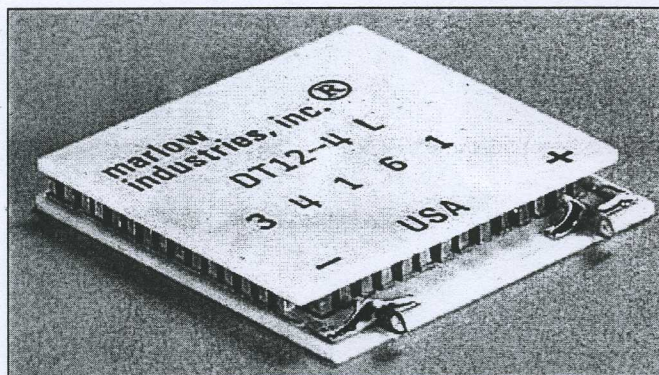
marlow industries inc.®

Thermoelectric Cooler

DT12-4

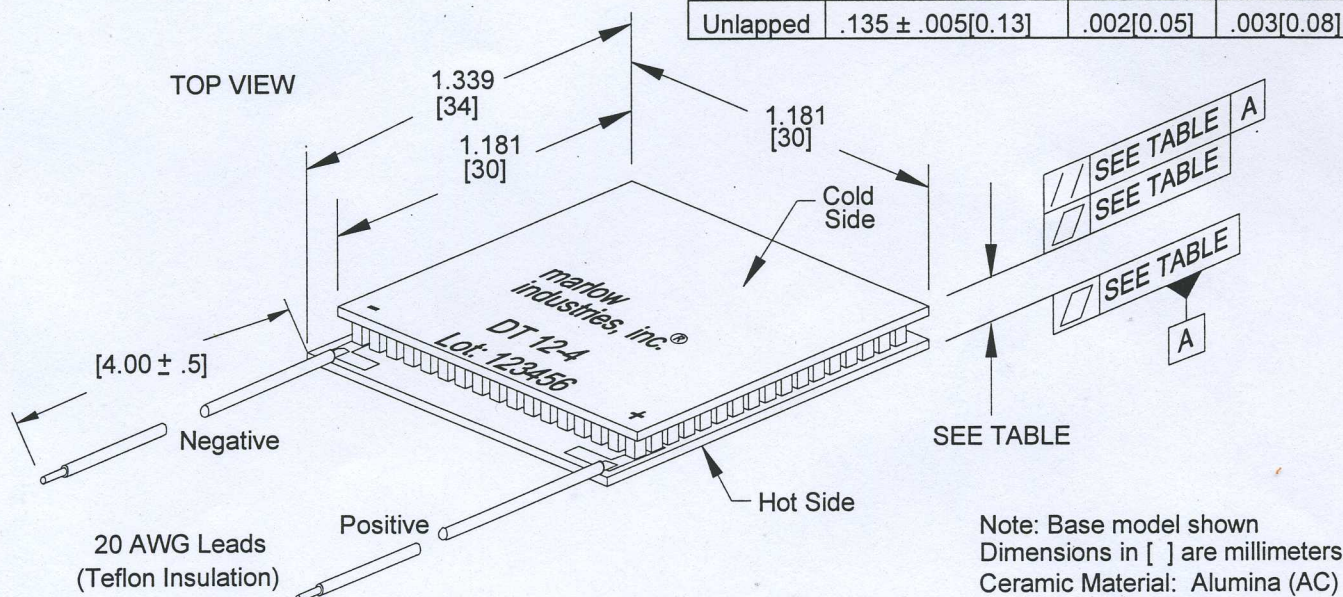
Performance Values

Hot Side Temperature (°C)	27°C	50°C
ΔT_{max} (°C-dry N ₂):	66	74
Q _{max} (watts):	36	39
I _{max} (amps):	3.7	3.7
V _{max} (vdc):	14.7	16.4
AC Resistance (ohms):	3.2	---



Mechanical Characteristics

	Height	Flatness	Parallelism
Lapped	.131 ± .001[0.03]	.001[0.03]	.002[0.05]
Unlapped	.135 ± .005[0.13]	.002[0.05]	.003[0.08]



Ordering Options

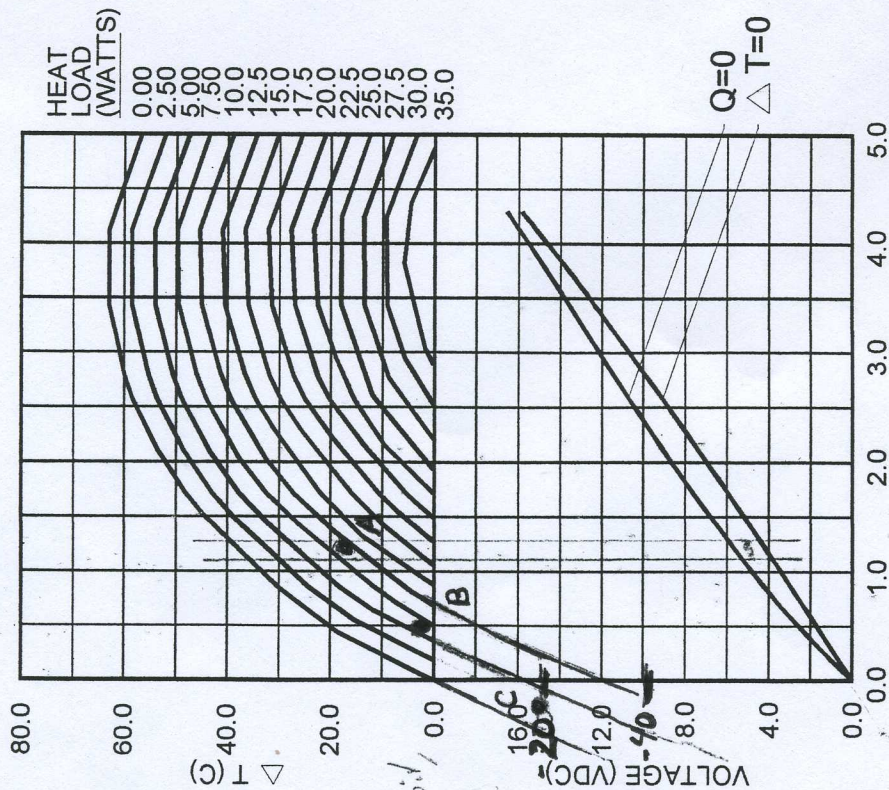
Model Number	Description
DT12-4-01	Base Model
DT12-4-01L	Lapped Model
DT12-4-01S	Sealed Model
DT12-4-01LS	Lapped and Sealed Model

Features

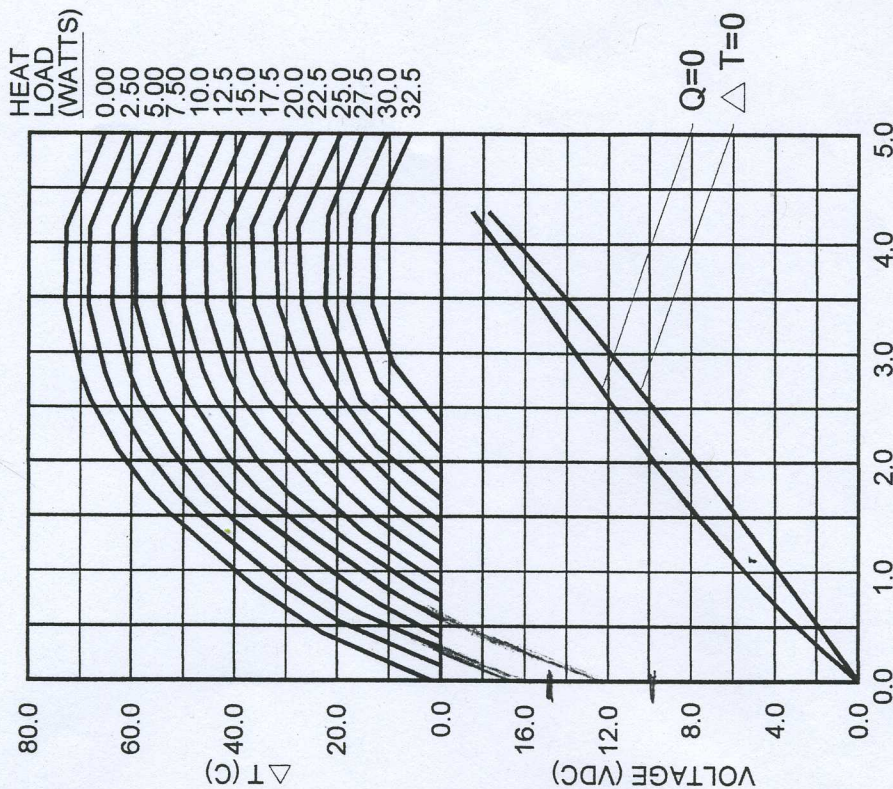
- Solid state reliability
- Built with high temperature solder with the ability to withstand higher assembly processing temperatures for short periods of time (<160°C)
- Superior nickel diffusion barriers on elements
- High strength for rugged environment
- Porched configuration for improved leadwire strength
- RTV sealing option available to improve reliability in condensing environment
- Lapped option available for multiple module applications.

DT12-4

Hot Side Temperature: 27°C



Hot Side Temperature: 50°C



opex points:	A	B	C
CURRENT (AMPS)	40C	20	0
amb	40C	20	0
cool air	30C	20	0
T _{PAX}	20C	20	20
P _{PAX}	12W	12	12
P _{total}	18	12	6

Q
AV_{1/2}

13.6 / 1.5 = 9.1 A

10.2 / 1.5 = 6.8 A

15.2 / 1.5 = 10.1 A

11.4 / 1.5 = 7.6 A

2 in series
@ 1.5A:

Zeigler-Nichols Tuning Method

The Zeigler-Nichols method for tuning PID controllers is presented in most control-engineering textbooks. The basic steps required to implement the Zeigler-Nichols method are as follows:

1. Set the integral and derivative coefficients to zero.
2. Gradually increase the proportional coefficient from zero until the system just begins to oscillate continuously. The proportional coefficient at this point is called the "ultimate gain", P_U . The period of oscillation at this point is called the "ultimate period", T_U . The ultimate period should be expressed in the same units used internally in the PID algorithm for calculations involving the time step, Δt . (The PIDSIm program uses seconds).
3. The Zeigler-Nichols suggested controller settings are as follows:

Control Action	Desired Performance	K_P	K_I	K_D
P	$\frac{1}{4}$ decay	$0.5 P_U$		
PI	$\frac{1}{4}$ decay	$0.45 P_U$	$0.54 (P_U/T_U)$	
PID	$\frac{1}{4}$ decay	$0.6 P_U$	$1.2 (P_U/T_U)$	$0.075 (P_U * T_U)$
PID	Some overshoot	$0.33 P_U$	$0.66 (P_U/T_U)$	$0.11 (P_U * T_U)$
PID	No overshoot	$0.2 P_U$	$0.606 (P_U/T_U)$	$0.10 (P_U * T_U)$

For our first example (5 second control interval) using the PIDSIm program, the ultimate gain was $P_U=110$, and the ultimate period was $T_U=20$. Using these values we can fill in the table.

Control Action	Desired Performance	K_P	K_I	K_D
P	$\frac{1}{4}$ decay	55		
PI	$\frac{1}{4}$ decay	49.5	2.97	
PID	$\frac{1}{4}$ decay	66	6.6	165
PID	Some overshoot	39.6	3.63	242
PID	No overshoot	24	3.33	220

For our second example (60 second control interval) using the PIDSIm program, the ultimate gain was $P_U=9$, and the ultimate period was $T_U=120$. Using these values we can fill in the table.

Control Action	Desired Performance	K_P	K_I	K_D
P	$\frac{1}{4}$ decay	4.5		
PI	$\frac{1}{4}$ decay	4.05	0.0405	
PID	$\frac{1}{4}$ decay	5.4	0.09	81
PID	Some overshoot	3	0.0495	108
PID	No overshoot	1.8	0.04545	118.8

Plug these numbers in and see how they work!

for PID transfer function

$$G_c(s) = K_p \left[1 + \frac{1}{T_i \cdot s} + T_d \cdot s \right]$$

P
I
D

$$= K_p \left(\frac{T_i \cdot T_d \cdot s^2 + T_i \cdot s + 1}{T_i \cdot s} \right)$$

Zeigler-Nichols method

- ① set derivative, + integral gain to zero
- ② turn up K_p until get steady oscillations
note $K_p = K_{crit}$, period of oscillation = T_{crit}

Z-N optima are

	K_p	T_i	T_d
P only	$0.5 K_{crit}$		
PI	$0.45 K_c$	$0.833 T_c$	
PID tight control	$0.6 K_c$	$0.5 T_c^*$	$0.125 T_c^{**}$
PID some overshoot	$0.33 K_c$	$0.5 T_c$	$0.33 T_c$
PID no overshoot	$0.2 K_c$	$0.3 T_c$	$0.5 T_c$

compared to reverse side $K_D = K_p T_D$

so $** T_D = 0.125 T_c \Rightarrow K_D = K_p T_D = .6 K_{cr} \cdot .125 T_{cr} = .075 K_{cr} T_{cr}$

$* T_i = .5 T_c \Rightarrow K_i = \frac{K_p}{T_i} = \frac{.6 K_{cr}}{.5 T_{cr}} = 1.2 \frac{K_{cr}}{T_{cr}}$